

Natural disasters detection using Machine Learning

Introduction

Natural disasters such as earthquakes, hurricanes, floods, and wildfires are unpredictable and can cause significant damage to property and losses of human lives. These events not only disrupt the functioning of societies but also require extensive efforts for recovery. Given their catastrophic impact, it is essential to have systems in place to predict and warn about these events, allowing people to prepare and take the necessary precautions. The Disaster Prediction System project is an ambitious attempt to develop a software system that utilizes advanced technologies such as machine learning, artificial intelligence, and data analysis to predict natural disasters. The Disaster Prediction System project is critical because it can help reduce the impact of natural disasters by providing early warnings and assistance to people in affected areas. In addition to predicting events, the system can alert relevant authorities and individuals through various channels such as mobile notifications, email alerts, and social media updates, enabling them to take necessary precautions and evacuate to safer locations. This information can help them allocate resources more effectively, reduce response times, and save lives. Historically, many people have suffered greatly due to the lack of proper disaster prediction and management systems. The inadequacy of accurate disaster predictions has often led to delays in evacuations and insufficient mitigation measures post-disaster. During the COVID-19 pandemic, for example, efficient steps could not be followed to prevent the further spread of the virus, revealing critical gaps in management systems. This paper addresses such issues by providing a

detailed review of existing procedures and techniques that can be employed during the pre-disaster and post-disaster periods to minimize losses as much as possible.

India is a vast country prone to a multitude of natural disasters. From the wrath of cyclones like Odisha in 1999 to 2004 Indian Ocean Tsunami to the recent landslides triggered by Cyclone Remal in the Northeast, the country has witnessed the destructive force of nature. According to NDMA, India faces major threats in the recent years.

- **Floods:** India experiences frequent floods, particularly during the monsoon season. Over 40 million hectares (12% of land) are prone to floods and river erosion. It is getting exacerbated by Glacial Lake Outburst Floods. ○ Examples: Floods in Bihar (2023), Assam (2022)

- **Cyclones and Storms:** India's coastline is susceptible to cyclones originating in the Bay of Bengal and the Arabian Sea. Of the 7,516 km long coastline, approximately 5,700 km is vulnerable to cyclones and tsunamis. ○ Examples: Cyclone Biparjoy (2023) and Cyclone Fani (2019).

- **Earthquakes:** India is located in a seismically active region, with several fault lines running across the country. 58.6% of the landmass is prone to moderate to very high intensity earthquakes. ○ Examples: Earthquake in Mizoram (2022), Earthquake in Sikkim (2011)

- **Droughts:** Prolonged dry spells and erratic rainfall patterns can lead to severe droughts, affecting agriculture and water resources. 68% of the cultivable area is susceptible to drought. ○ Examples: Drought in 66% of Maharashtra (2024).

- **Landslides:** Hilly and mountainous regions in India are prone to landslides, especially during heavy rainfall or earthquakes. India is

considered among the top five landslide-prone countries globally. ○ Examples: Landslides in Himachal Pradesh (2023), Landslides in Manipur (2022).

■ Heat Waves: Rising temperatures and prolonged periods of high heat can lead to life-threatening heat waves. ○ Examples: Heat waves across India (2022, 2023, 2024). ● The country reported 280 heat wave days from 11th March- 18th May, 2022.

■ Forest Fires: Dry conditions and human activities can contribute to forest fires, causing environmental damage and air pollution. Based on the forest inventory records, 54.40% of forests in India are exposed to occasional fires ○ Examples: Forest fires in Himachal Pradesh (2024) and Forest fires in Goa (2023).

■ Industrial and Chemical Accidents: India's growing industrial sector and improper handling of hazardous materials can lead to industrial and chemical accidents. ○ Examples: Chemical spill in Surat (2023), Industrial fire in Mumbai (2024)

The field of AI and ML applications in disaster prediction has witnessed a fascinating evolution. Early research efforts focused on harnessing these technologies for earthquake prediction, with significant advancements in model development. This initial push laid the groundwork for exploring AI/ML's potential in other areas of disaster management

Applications of ML in Disaster Management

- **Earthquake Prediction and Impact Analysis** ML models use seismic data to detect early patterns that precede earthquakes. Support vector machines and neural networks classify tectonic movements.

Regression models estimate intensity and potential impact. These systems are enhanced by integrating geospatial data and historical seismic records.

- **Flood Forecasting** Algorithms analyze rainfall, river flow, and weather data to predict floods. LSTM and gradient boosting models provide real-time forecasting, allowing timely evacuation and floodplain management. GIS integration allows for visualization and mapping of flood risks.
- **Wildfire Detection** CNNs analyze satellite imagery to detect smoke and fire-prone areas. ML models forecast wildfire spread using meteorological data, vegetation indices, and terrain models. Early detection systems can now alert authorities within minutes of ignition using real-time image feeds and anomaly detection.
- **Landslide Monitoring and Prediction** Using terrain slope, rainfall, and soil composition data, ML models like Random Forests and Gradient Boosting predict landslide-prone areas. Unsupervised clustering helps in categorizing regions with similar vulnerability profiles.
- **Cyclone and Hurricane Tracking** ML aids in tracking the intensity, trajectory, and landfall timing of cyclones. Historical storm data is combined with real-time satellite feeds and wind speeds to improve forecasting models. Decision support tools recommend evacuation zones and resource pre-deployment.
- **Disease Outbreak Prediction** ML models analyze travel data, social media trends, and historical health records to predict outbreaks. Logistic regression, support vector machines, and ensemble methods are used to estimate infection rates. These models offer early detection, often days ahead of official reports.

- **Contact Tracing and Transmission Mapping** Using graph-based models and mobile data, ML helps trace contacts and predict possible infection chains. Algorithms compute infection likelihood scores for individuals based on proximity and exposure time. Apps from Apple, Google, and national governments have used these models to notify users of potential exposure.
- **Vaccine Distribution Optimization** Reinforcement learning and optimization algorithms help manage logistics for vaccine delivery. Factors include cold storage capacity, population density, regional demand, and transportation delays. These models reduce wastage and improve coverage.
- **Diagnosis and Symptom Classification** ML models have been used in symptom checking apps and virtual assistants. Natural language processing (NLP) and decision trees help triage patients, reducing pressure on hospitals. CNNs analyze X-rays and CT scans for diagnosis of diseases like COVID-19.
- **Hospital Resource Management** Predictive models forecast ICU demand, oxygen requirements, and medicine stocks. These insights assist in proactive planning and avoid shortages during peak waves.

Disaster Prediction Models in Machine Learning:

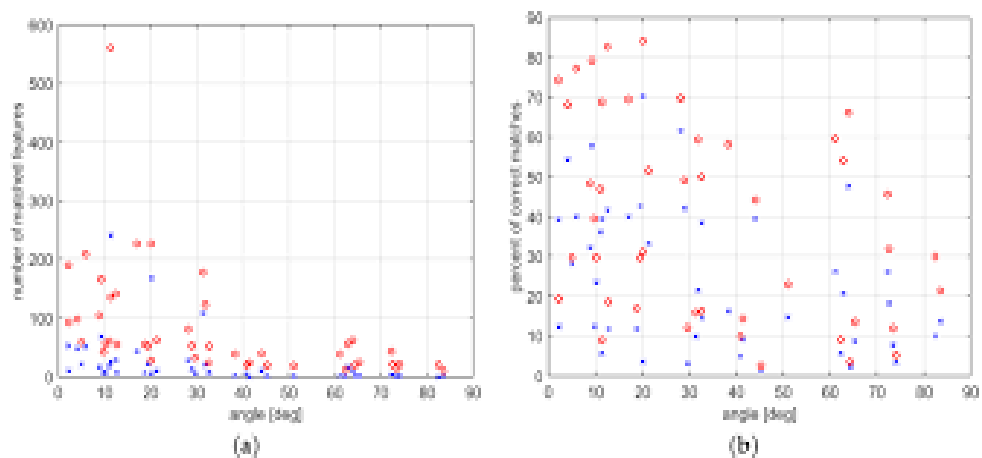
○ SMV MODEL

SVM operates by searching for optimal classification hyperplanes in both linearly separable and inseparable cases, utilizing structured risk minimization to construct the optimal hyperplane in

the attribute space, ensuring the classifier achieves the global optimum and meets the expected risk at a certain upper bound . Kernel functions are crucial in the effectiveness of SVM algorithms. These functions enable SVM to handle non-linearly separable data by mapping it to a higher-dimensional space, where linear separation becomes feasible [

Analysis of SVM Model Results

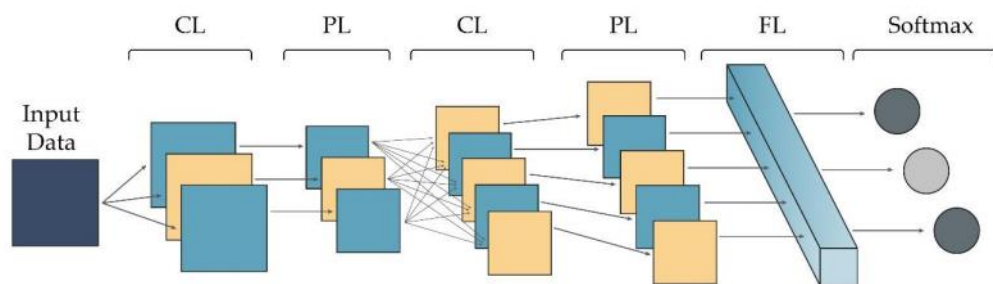
In the evaluation of the SVM model for tsunami prediction, several vital metrics were employed to assess its performance. The model demonstrated an accuracy of 65.61%, precision of 70.59%, recall of 19.67%, F1-score of 30.77%, and a Receiver Operating Characteristic (ROC) Area Under the Curve (AUC) of 62.15%. These results illuminated the SVM model's strengths and limitations in its application to tsunami prediction.



- **CNNs**
- The CNN architecture is based on convolutional layers (CL). In these layers the data is propagated and applied with tabular multiplication of $n \times m$ tabular filters also called kernels, where n equals m in most cases. This process produces different representations of the input data based on the filter applied to

them. In each representation various features are uncovered which are projected in feature maps that quantify the stimuli produced by each filter of the convolutional layer . As a result of the convolution, multiple unique transformations of the input data are generated. The number and size of kernels that are used in the convolution are key for the performance of the overall network. After the convolution step, the data is passed through a pooling layer (PL) that groups the convolution results and retains only the most important ones to the network, by keeping the maximum, minimum or average of each part of the data. This entire process is repeated several times depending on the depth of the network before the data is flattened in a one-dimensional space in order to be fed in a fully connected layer (FL) which handles the classification. The softmax layer converts the data into a probability distribution. Figure shows a CNN architecture. It is important to mention that each CL in the network provides a level of abstraction in the feature extraction process.

These models assist in identifying flood-prone areas and assessing damage extent.



Case Studies:

1. COVID-19 and AI-Powered Forecasting (BlueDot)

In late December 2019, BlueDot, a Canadian AI company, identified an unusual cluster of pneumonia cases near a market in Wuhan, China. Using Natural Language Processing (NLP) and machine learning models, it analyzed global news in 65 languages, airline ticketing data, and animal disease networks. BlueDot was able to issue an early warning on December 31st—days before WHO's official statement.

Model Used: NLP + Decision Trees

Impact: Enabled preemptive alert systems and influenced public health surveillance networks.

2. NASA Fire Information for Resource Management System (FIRMS)

NASA's FIRMS system integrates machine learning with satellite data to monitor wildfire activity worldwide. ML models process MODIS and VIIRS imagery to detect heat anomalies and predict fire spread.

Model Used: CNNs on satellite thermal bands

Impact: Real-time alerts for fire-prone regions and improved firefighting logistics.

3. India's FloodNet: Urban Flood Prediction

FloodNet is an ML-powered sensor network deployed in Indian cities like Hyderabad. It predicts urban floods by combining rainfall data, drainage flow, and soil saturation levels. The system uses LSTM networks for real-time flood forecasting.

Model Used: LSTM + Ensemble Models

Impact: Enabled early warnings and reduced damage in flood-prone urban pockets.

4. Taiwan CDC: Smart Pandemic Surveillance System

During the COVID-19 outbreak, Taiwan's CDC implemented an ML-based surveillance system that integrated travel history, health records, and quarantine monitoring via mobile phones. High-risk individuals were flagged and isolated proactively.

Model Used: Logistic regression + geospatial models

Impact: Controlled the spread efficiently with minimal lockdown disruptions.

5. AlphaFold for Vaccine and Drug Discovery

DeepMind's AlphaFold leveraged deep learning to predict 3D protein structures. During the COVID-19 pandemic, it helped researchers understand the SARS-CoV-2 virus structure, accelerating vaccine development.

Model Used: Deep Neural Networks for structure prediction

Impact: Reduced drug discovery timelines from years to weeks.

6. Cyclone Forecasting in the Philippines

The Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) implemented ML models to predict cyclone paths. Decision trees and ensemble models combined meteorological data to improve accuracy.

Model Used: Random Forests + SVM

Impact: More accurate evacuation plans, saving thousands of lives during Cyclone Rolly.

7. Predictive Healthcare in South Korea

South Korea used ML models to estimate ICU occupancy and oxygen demand across hospitals during COVID-19. Dashboards were built in collaboration with private tech companies.

Model Used: Gradient Boosting + Time-series analysis

Impact: Enabled dynamic reallocation of medical resources and reduced case fatality rates.

Citations

1)Drishti IAS. (n.d.). *Making India disaster resilient*. Retrieved April 22, 2025, from <https://www.drishtias.com/daily-updates/daily-news-analysis/states-demand-to-declare-lightning-as-a-natural-disaster>

2)Predictive Analytics in Disaster Prevention: Machine Learning Models for Early Warning Systems:<https://publicsafety.ieee.org/topics/predictive-analytics-in-disaster-prevention-machine-learning-models-for-early-warning-systems>

3)Case study on covid 19- <https://bluedot.global/>

4) New Indian Express. (2024, July 22). *IIT-BBS' AI model to accurately measure intensity of cyclones in NIO basin*. Retrieved April 22, 2025, from <https://www.newindianexpress.com/states/odisha/2024/Jul/22/iit-bbs-ai-model-to-accurately-measure-intensity-of-cyclones-in-nio-basin>

5) Comparative Analysis of SVM and RF Algorithms for Tsunami Prediction: A Performance Evaluation Study- <https://www.bright-journal.org/Journal/index.php/JADS/article/download/159/135>

6) Chamola, V., Hassija, V., Gupta, S., Goyal, A., Guizani, M., & Sikdar, B. (2020, December 15). *Disaster and pandemic management using machine learning: A survey*. IEEE internet of things journal. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8768997/>

7) Ding, X., Shang, B., Xie, C., Xin, J., & Yu, F. (2025, February 23). *Artificial Intelligence in the COVID-19 pandemic: Balancing benefits and ethical challenges in China's response*. Nature News. <https://www.nature.com/articles/s41599-025-04564-x>

